[Paper 5] Paper "CosmoRun" & "CGM" - 1e12q Run To Study Absrp. Lines

This page presents the outline, ongoing efforts, results, and future plans for the AGORA Paper "CGM" (being co-led by Hummels, Roca-Fabrega, etc).

Contents

Important recent changes that require your attention are in red color.

(last revision: 03/09/2021)

1. Project Outline

	Paper "CosmoRun" + Paper "CGM"
GOALS	• **Paper "CosmoRun" :** Introduce ~1e12 Ms halo run, emphasis on the calibration process and how it differs from previous attempts like Aquila, discuss basic properties of halos, using the cosmological run data as early as z=3.5 or 3 • **Paper "CGM" :** Study absorption lines, focus on CGM properties that are not resolution-dependent (e.g., high ions), using the cosmological run data as early as z=2
LEADERS	• Cameron Hummels (Caltech) • Santi Roca-Fabrega (UCM)
PARTICIPANTS	• Interested code representatives: **Code** — **Contacts** ART-I — Dekel, Mandelker, Primack, Roca-Fabrega, Strawn ART-II — (Feldmann) CHANGA — Powell, Quinn ENZO — Barrow, Hummels, Kim (Ji-hoon), Kim (Ki-won), O'Shea, Shin, Wise GADGET — Nagamine, Shimizu GASOLINE — (Madau, Shen, Wadsley, Wissing) GEAR — Hausammann, Revaz GIZMO — Dong, Hopkins, Hummels, Lupi, Velazquez ("favorite" fbck is their standard mechanical fbck, GALSF_FB_MECHANICAL in here, *not* FIRE-2) RAMSES — Agertz, Roca-Fabrega, Teyssier (Code leaders as of Feb. 2020 / ICs help: Buehlmann, Hahn, Simpson / Analysis help: Chakrabarti, Fumagalli, Hummels, Kim, Madau, Nagai, Roca-Fabrega, Smith, Turk) • Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - V (subgrid physics) - IX (Characteristics of Cosmo. Galaxies) - X (Outflows)
	• We will compare the run with observed absorption lines data from e.g., COS-Halos at low redshift (MW-like halo masses, 1.5 < z < 3.5, webpage, Tumlinson et al. 2013), KBSS at higher redshift (Rudie et al. 2012, Steidel et al. 2014), CASBaH for NeVIII (Burchett et al.), or emission-line galaxies at z ~ 0.7 (ELG, Prochaska et al.). Observers will certainly be interested.

<table>
<tr><td>SCIENCE CASE</td><td>• More recent CGM-related articles to consider: Peeples et al. 2019 (Astro2020 white paper), Lehner et al. 2019, Ng et al. 2019
• We may later run one or two MHD runs for comparison.
• We may consider running the sim down to z=0 and compare the metallicity distribution not just in the CGM, but on the disk. This will lead us to a separate Paper "Metallicity", or a section in Paper "CGM".</td></tr>
<tr><td>PRE-PRODUCTION & MID-PRODUCTION DISCUSSION

1ST MEETING (04/24/2017)

2ND MEETING (07/06/2017)

3RD MEETING (08/12/2017)

4TH MEETING (06/11/2018)

5TH MEETING (08/11/2018)

6TH MEETING (12/10/2018)

7TH MEETING</td><td>• Pre-production discussion/to-do's:
<table>
<tr><th>Item</th><th>To-do</th></tr>
<tr><td>Initial condition</td><td>• AGORA quiescent 10^{12} Ms halo at z=0. (More information about the halo at this link.) This early-forming halo will be an interesting object even at high z.
• A MUSIC configuration file for an ellipsoidal IC with levelmax=12 and a Lagrangian region of $4R_{vir}$ is created by Buehlmann and Hahn (z=0 snapshot of a DM-only test with levelmax=11, movie).
• In fact, they now have built a full library of AGORA ICs with different choices of Lagrangian regions for progenitors at different redshifts (to be formally announced soon; example plot).
• This IC is designed to give particle masses m_{DM} = 2.8×10^5 Ms and m_{gas} = 5.65×10^4 Ms, close to the resolution adopted in our disk comparison paper (formerly Paper "4"; m_{gas} = 8.59×10^4 Ms, dx = 80 pc). We assume that the disk comparison paper found a reasonable way to match simulation resolutions across codes (SPH vs AMR, PLM vs PPM), and will simply piggyback on it.
• The initial metallicity should be set to 10^{-4} Zs everywhere.
• We may consider using the 10^{13} Ms halo to inspect CGM at higher redshift, e.g., z=2 or 4.</td></tr>
<tr><td>Target redshift</td><td>• z=2 (initial target)
• z=0 or 1 (secondary target if we later decide to run the sim down to lower redshift)</td></tr>
<tr><td>Common physics (1): Stellar feedback</td><td>• We will run simulations with two modes of feedback:
- (1) "Common" stellar feedback prescription that is identical to our disk comparison paper (formerly Paper "4"). This is thermal-only feedback.
- (2) "Favorite" stellar feedback prescription(s) of each code group. For "favorite" prescriptions, we ask to provide *at least one* simulation with a mainstream prescription (i.e., thermal+kinetic feedback). Beyond that, you are free to also provide whatever extreme you want to do (e.g., MHD, CR).

• The "favorite" feedback prescriptions will need to be first calibrated across codes to produce similar properties (e.g., stellar masses), and to eject metals into CGM. For this, we will carry out a 2-step calibration as follows:
- (0) Calibration Step #0: We will first run the isolated disk IC (identical one from our disk comparison paper, formerly Paper "4"; see also its Workspace that includes the IC) with both "common" and "favorite" feedback recipes. Each group will tweak its "favorite" recipe to give a stellar mass of ~10^9 Ms produced in the first 1 Gyr -- within a factor of 2 difference is probably okay. (See Figure 26 of disk comparison paper, formerly Paper "4"; how about SFR? or velocity dispersion?) Meanwhile check if your old run for our disk comparison paper with the "common" feedback recipe produced a similar stellar mass. For this step, you may need to talk to the person who ran the simulation for the disk comparison paper, to acquire the IC, and the parameter set we settled on only after multiple trials and errors. This step has been largely cleared by all participating codes.
- (1) Calibration Step #1: We will then run the cosmological IC of the 10^{12} Ms halo only with radiative gas cooling (no star formation), and compare the gas mass distribution in e.g., the density-temperature PDF at z=4. This step is to make sure that the required Grackle v3.1.1 has been properly and fully incorporated into each code, and that the cooling works properly. Since we are doing this at much earlier redshifts, please submit your data as frequently as possible (e.g., as early as z=10, 8, ... is fine) so we can monitor the basic diagnostics such as PDFs, and the emergence of clumps for Paper "Clumps". This step has also been largely cleared by all participating codes.
- (2) Calibration Step #2: We will then run the cosmological IC of the 10^{12} Ms halo with star formation but no feedback, and compare e.g., the stellar mass evolution by z=7. This step is to make sure that the common star formation recipes are properly implemented (and calibrated if a different approach has been adopted) across different codes. This step has also been largely cleared by at least 4 participating codes as of early 2020.</td></tr>
</table>
</td></tr>
</table>

(01/23/2019)

8TH MEETING (03/06/2019)

9TH MEETING (04/09/2019)

10TH MEETING (05/09/2019)

11TH MEETING (06/14/2019)

12TH MEETING (07/19/2019)

13TH MEETING (08/10/2019)

14TH MEETING (08/23/2019)

15TH MEETING (10/01/2019)

16TH MEETING (10/29/2019)

17TH MEETING (11/14/2019)

18TH MEETING (12/13/2019)

19TH MEETING (02/10/2020)

20TH MEETING (03/18/2020)

21TH MEETING (06/25/2020)

22TH MEETING (07/16/2020)

23TH MEETING (07/23/2020)

24TH MEETING (07/30/2020)

25TH MEETING (08/18/2020)

26TH MEETING (08/28/2020)

27TH MEETING

<table>
<tr><td></td><td>– (3) Calibration Step #3: We will then run the cosmological IC of the 10^{12} Ms halo with star formation and "favorite" feedback, and compare e.g., the stellar masses, velocity dispersion, etc. across codes at z=4. We will run with "common" and "favorite" feedback recipes to check if the halo has a similar cooling flow, accretion, etc. Each group will make sure that its "favorite" recipe gives a stellar mass of ~ 10^9 Ms at z=4 (notice the change from ~ 10^8 Ms to ~ 10^9 Ms as of 07/2019), and decide which one is their preferred entry.
– [3-A] This ~ 10^9 Ms value is based on M_{halo} – M_* relations in recent SHAM studies. For our 1.7×10^{12} Ms halo at z=0 (or ~10^{11} Ms at z=4 according to the ordinary growth history found in SHAMs, but ~2×10^{11} Ms at z=4 according to the accretion history of this particular early-forming AGORA halo), the stellar mass is expected to be ~ $0.5-1\times10^9$ Ms (see Primack's justification here -- file revised on 08/10/2019).
– [3-B] Note that the stellar mass of a typical 10^{12} Ms halo at z=0 has 1-sigma scatter centered at 0.5, 1.0, 1.5×10^8 Ms at z=4 in Moster et al. 2018 (Fig.10), Behroozi et al. 2018 (Fig.16), Rodriguez-Puebla et al. 2017 (Fig.23), respectively. For more discussion, see "General Issues" below.</td></tr>
<tr><td>Common physics (2):
Pressure floor</td><td>• We will adopt the prescription used in our disk comparison paper (formerly Paper "4").</td></tr>
<tr><td>Common physics (3):
Star formation</td><td>• We will use the common prescription established in our disk comparison paper (formerly Paper "4"). And as of 11/2019, we are testing and switching to n_thres = $1 H/cm^3$ instead of $10 H/cm^3$ (see more discussion below). Each group will get a stellar mass of ~ $1-5\times10^9$ Ms at z=4 only by tweaking the feedback recipe. Note that the minimum stellar particle mass should be 6.1×10^4 Ms (the value was $m_{gas} = 8.6\times10^4$ Ms in Paper "4"; should have been 5.65×10^4 Ms – error noticed by Tom/Santi in 08/01/2020).</td></tr>
<tr><td>Common physics (4):
Cooling & UVB</td><td>• We will adopt the Grackle equilibrium cooling used in our disk comparison paper (formerly Paper "4") (i.e., Grackle v3.1.1 with self_shielding_method = 0 and primordial_chemistry = 0, i.e., the table CloudyData_UVB=HM2012_shielded.h5, instead of the default CloudyData_UVB=HM2012.h5).

• Though during our comparison Revaz and Hausammann have identified a bug in Grackle v3.1 or below affecting the high-z era, we decided to not use Grackle v3.1.1 that contains the fix, in the current runs, in order not to introduce any more variables and confusion. Since then, Hausammann's test showed that indeed the correct Grackle v3.1.1 introduces non-negligible effect in the results. Therefore, as of 08/2019, we will migrate to the v3.1.1 in the next generation of runs (see "TIMELINE AND MILESTONES" below for our action plan). The presently-running calibration runs may still use Grackle v3.1 for the calibration step #2.
• Other Grackle-related parameters to check include: UVbackground_redshift_on = 15, UVbackground_redshift_off = 0, UVbackground_redshift_fullon = 15, UVbackground_redshift_drop = 0, SolarMetalFractionByMass = 0.02041.
• For high ions we want to focus in Paper "CGM", self-shielding may not cause much difference. But for star-forming clumps we want to focus in Paper "Clumps", self-shielding will potentially play an important role.</td></tr>
<tr><td>Code-dependent physics:
Metal diffusion in SPH</td><td>• Each group will decide what they want to do. For example, GASOLINE/CHANGA/GIZMO have an explicit metal diffusion scheme (e.g., Shen et al.), while GADGET/GEAR have an "implicit" scheme that is based on kernel-weighting over nearest neighbors.</td></tr>
<tr><td>Softening, smoothing, and refinement strategy</td><td>• We will adopt a similar strategy chosen for our Flagship paper and our disk comparison paper (formerly Paper "4"):
<table>
<tr><td></td><td>Chosen values</td></tr>
<tr><td>levelmax in MUSIC parameter</td><td>12</td></tr>
<tr><td>Mass resolution in the finest level</td><td>$m_{DM} = 2.8\times10^5$ Ms
m_{gas} = 5.65×10^4 Ms
(3.4×10^5 Ms in DM-only run)</td></tr>
<tr><td>SPH gravitational softening length
(= eps_{grav} = equivalent Plummer softening length)
in the highest-resolution region</td><td>800 comoving pc until z=9,
80 proper pc afterwards
(see note 1 below),
the same softening length used for all particle types</td></tr>
<tr><td></td><td></td></tr>
</table></td></tr>
</table>

(09/09/2020)

28TH MEETING (09/29/2020)

29TH MEETING (11/18/2020)

30TH MEETING (12/16/2020)

31TH MEETING (01/13/2021)

32TH MEETING (02/09/2021)

33TH MEETING (03/09/2021)

SPH minimum smoothing length	0.2 * eps_{grav}
AMR finest resolution cell size	163 comoving pc, i.e. max refinement level = 12 (see note 2 below)
AMR refinement criterion	particle & gas overdensity of 4

• While recent papers argue that pc or sub-pc resolution seems to be needed to match observation (e.g., Hummels, Katz, Simons), our 80 pc resolution should be enough to set the stage for code comparison, and focus on CGM properties that are not resolution-dependent (e.g., high ions).

Output strategy

• We recommend 200 snapshots starting at scale factor of 0.062 ~ z=15, equally spaced timesteps in log(expansion parameter), Δlog(a) = | log(384/2013)/200 |, from a=0.062 (z~15) to a=0.325 (z~2). Plus, redshift snapshots at z = 15.0, 14.0, 13.0, 12.0, 11.0, 10., 9.0, 8.0, 7.0, 6.0, 5.0, 4.0, 3.0. The explicit list of outputs can be downloaded **here** (thanks to Roca-Fabrega).
• You are of course encouraged to store more snapshots on our large NERSC disk space. FYI, see **this link** for a default output strategy in AGORA -- which is different from what we adopted here.

Common analysis

• Hummels has provided (08/2017) and updated (08/2018) an **ipython notebook** that explains how to use yt+**Trident** on datasets from various codes. Here the notebook should in principle work with both yt-3.5 and yt-4.0 (the latter still experimental as of 01/2019; see below about demeshened yt-4.0).
• Trident is fully functional with "demeshened" version of yt (i.e., **yt-4.0** branch but still not officially released; see another **ipython notebook** for how to install yt-4.0). As of 06/2018, yt-4.0 enables particle-based codes to be treated more natively by not depositing to grids (see **this link** **this link**, **this link**; see also **this link** about Trident v1.1 as of 11/2017 with some preliminary ideas for "demeshened" version of yt and Trident).
• The yt(-4.0)+Trident pipeline has been widely field-tested by now for a number of codes (ART-I/ENZO/CHANGA/GADGET/GIZMO) by Hummels, Strawn, Butsky, etc, as of 01/2019.
• Strawn volunteers to incorporate his yt+Trident script to produce the absorption lines, HI covering fraction, etc. Strawn and Nagamine have worked to make GADGET-3 to fully work with Trident, and gone through most of the entire CGM pipeline with GADGET-3. Some remaining issues with SPH+Trident have to do with how to handle lightcone-style data with multiple redshifts, which shouldn't affect our analysis for Paper "CGM".
• Strawn also volunteers to generate ~400 random sightlines and heat maps on e.g., radius - column density plane into the **agora-analysis-script** repository. An example can be found **here** or in **Section 2** using ART-I and ENZO data; see "Key plots to make" below.
• Analysis step could potentially be quite straightforward. Hummels has several scripts that he is willing to share with the community that enable one to do radial profiles (i.e., column density of some ion versus impact parameter) pretty easily, and then we can compare against the COS-Halos results.
• One may expect that *all* of the groups are going to be off in trying to match the COS-Halos results at some level, so no one should feel bad about not matching them. It will be instructive to understand why certain groups match some ions and other groups match other ions.
• We encourage code groups to submit multiple snapshots with different "favorite" feedback prescriptions. This will produce "brackets" or "bands" in the plots, rather than points. Ideally these "bands" of different codes will overlap with one another, and with the observational data.

Key plots to make

• Roca-Fabrega will initially make an analysis pipeline for (see **Section 2** for example figures):
  - (1) Density-temperature phase diagram (color code weighted by mass or by volume)
  - (2) Radial profiles of gas density, temperature, metals or mass-weighted metallicity
  - (3) Vertical profile of outflow velocity (weighted by mass or by metallicity)
• Strawn will initially make an analysis pipeline using his yt+Trident experience for (see example **here** or in **Section 2**):
  - (1) Absorption lines for high ions (CIV, NV, NVI, OV, OVI, OVII, OVIII, NeVIII, MgX and possibly more, but let us first focus on OVI and NeVIII that are less resolution-dependent considering our poor resolution in the CGM)
  - (2) Radial profiles of metal species, line widths of the lines (kinematics; proposed by Burchet)
  - (3) Inflow/outflow rates (mass fluxes through radial shells, or kinematics along the lines of sight, structure of lines)
  - (4) HI covering fraction (how much of a region out to ~100 kpc is covered by a certain HI column density cut, this may not give accurate results due to resolution, but that is already a good science),

		HI clumping factor – (4) Column density profiles (as a function of impact parameter), line absorption profiles (as a function of wavelength) (proposed by Faerman) – (5) OVI produced by photoionization vs. collisional ionization as a function of redshift (?; to be discussed)
	Obs. counterpart	• COS-Halos or CASBaH (see "SCIENCE CASE" above for references; KBSS are for higher mass halos)
TIMELINE AND MILESTONES	• Before and during the Aug. 2017 Workshop: Make cosmological IC ready, learn how to use **Trident** on yt. • Before and during the Aug. 2018 Workshop: Calibration step #1 (see above), finalize runtime parameters and simulation strategy. • Online meeting on Dec. 10, 2018: Status checks on each group's calibration step #2, code leaders asked to write about their "favorite" feedback/cooling of their dataset entries in the "3. Physics" section, focus on the parts that are different from the disk comparison paper, we will refer to this section in the Paper "Clumps" later. • Online meeting on Jan. 23, 2019: Status checks on each group's calibration step #2, discussion on preliminary plots made at z=10 and 6, discussion on possible causes for inter-code differences (metallicity, UVB, self-shielding, etc.), task assignments for analysis tools. • Online meeting on Mar. 06, 2019: Status checks on each group's calibration step #2, discussion on preliminary plots made at z=10 with the new initial metallicity of 10^{-4} Zs, discussion on possible issues in each code and the high SFR overall, Strawn's script. • Online meeting on Apr. 09, May 9, Jun. 14, Jul. 19, 2019: Status checks on each group's calibration step #2, discussion on common cooling, self-shielding, and SF prescription. • Before and during the Aug. 2019 Workshop: Common analysis tools almost ready and being used to make key figures proposed in the Aug. 2018 Workshop. Start writing the "5. Results and Discussion" section. • Shortly after the Aug. 2019 Workshop: End of calibration step #2 at z=4. Code leaders are asked to run with Grackle v3.1.1 down to z=3, along with a DM-only run (the latter should be very easy). Code leaders are also asked to run a short simulation without star formation and feedback down to z=8 (see more about the justification of this run by searching for "turnaround density" below). Code leaders are asked to continue to submit updated datasets, as early as z=10 down to 3. • By 12/2019: Results at z=3 absolutely needed by 12/2019. We will let the code run until z=2 for Paper "CGM" while we work on Paper "CosmoRun". First full version of Paper "CosmoRun" ready. • Early 2020: Paper "CosmoRun" submitted. • Aug. 2020 Workshop: With the completion of Paper "CosmoRun", we will quickly finish up Paper "CGM" and relaunch/reinitiate the 9 Science-oriented Working Groups. For more information, please see our discussion in the 2019 Workshop in **this link** (Section 3-(3)).	
OTHER LINKS	• **Trident download and documentation** • **Trident method paper (Hummels et al.)** • **WG III Workspace (AGORA Cosmological ICs)** • **WG IX Workspace (Characteristics of Cosmological Galaxies)** • **WG X Workspace (Outflows)**	

[note 1] If we had followed the same exact prescription in the flagship paper (see **this link**), the SPH softening length should have been $0.5*4*R_{200}/\mathrm{sqrt}(N_{200})$, from Mike Kuhlen's suggestion of 50% of epsilon_opt found in Power et al. 2003:

$\sim= 0.5*4*(1.7e12*2e33)^{1/3} * [200*(4*3.14/3)*1.88e-29*(0.702)^2]^{-1/3} / 3.086e18 / (1.7e12/3.4e5)^{1/2} = 220$ pc

However, we want to impose a better resolution than this: 80 pc resolution at z~2-3, the resolution in our disk comparison paper (formerly Paper "4", see **this link**) and the scale at which our subgrid physics models are calibrated.
Therefore, the suggested softening parameter is:

-> epsilon = 80 pc for the highest-resolutuion region
-> epsilon = 80 pc * $8^{[(\mathrm{level\_max-level\_here})/2]}$ for the lower resolution regions because epsilon goes as ~ 1/sqrt(N) ~ $\mathrm{sqrt}(m_{\mathrm{particle}})$

[note 2] 60 Mpc/h/$2^7/2^{12}$ = 163 pc, where level = 0 is the root grid with 2^7 cells, i.e. levelmin = 7 in the MUSIC parameter

2. Results

Below we post the first batch of composite figures we agreed to produce for Paper "CosmoRun" and Paper "CGM". This is hardly the final result of the comparison, so note that some figures may change significantly as we identify and fix various issues. Reload the images to make sure you are looking at the most recent ones. Please also read Code Notes below carefully before drawing any conclusion. We also kindly remind you of the **internal policy** regarding the disclosure of AGORA simulation results. The **agora-analysis-script** repository is being updated for the common yt script used to carry out the analysis presented here (script-Paper4-disk-new.py).

(1) Paper "CosmoRun" Draft on Overleaf

- Overleaf link: **https://www.overleaf.com/3476292697whprtcfpgytw**
- Please do not edit this version now (starting Mar. 23, 2021), but send your last-minute feedback to Santi and Ji-hoon directly.

(2) Paper "CGM" Draft on Overleaf

- Overleaf link: **https://www.overleaf.com/18527533qnjhwwxkydrx**

(3) Calibration of Feedback Prescriptions

	Data Locations, Visualizations, Metrics to Compare
Calibration Step #1	• Composite figures and plots for the calibration step #1 are being collected at: **https://drive.google.com/drive/folders/1QVCcksel9Q37J6Izmed2LFr-ss5AyBkC** • Code Notes: – (1) ART-I: A different cooling and heating schemes were used (i.e., not Grackle, but calibrated using an isolated disk to be the same). – (2) GEAR: Only "favorite" feedback is calibrated here in step #1. – (3) GIZMO: "Favorite" fbck is their standard mechanical fbck, GALSF_FB_MECHANICAL in **here**, not FIRE-2.
	• Composite figures and plots for the calibration step #2 are being collected at the link below, but more recent plots are also being circulated via emails within code leaders. **https://drive.google.com/drive/folders/1DGR70Z71K6QucQpo4Vjn2x6I04znqE__** • Composite plots made with yt+Trident using ART-I and ENZO datasets: **example**. See code notes below for possible remaining issues with the yt+Trident analysis pipeline. • General Issues: – (1) At z~4, some code groups see M_* higher than expected by SHAM (>3x10^8 Ms) for its halo mass this time around (~2x10^{11} Ms at z=4). As of 03/2019, multiple code groups are seeing, roughly, M_* ~ 2x10^9 Ms, M_{gas} ~ 3.5x10^9 Ms, M_{halo} ~ 3x10^{10} Ms at z=6, and M_* ~ 2x10^8 Ms, M_{gas} ~ 2x10^9 Ms, M_{halo} ~ 1.5x10^{10} Ms at z=10. We think that this is probably expected given the "traditional" feedback implementations we have adopted. – (2) As of 07/2019, we decided to raise our stellar mass benchmark from ~ 10^8 Ms to ~ 10^9 Ms. Check the "Common physics (1):Stellar feedback" above to learn more about this. The higher than expected stellar mass might have been caused by the difference in cosmology adopted (WMAP7 for AGORA vs. Planck15 for SHAM; see more info about the adopted

Calibration Step #2	cosmology at this link). At higher redshift, stellar masses in halos may depend more sensitively on cosmological parameters than at lower redshift. – (3) As of 10/2019, two issues were raised. (3-1) First, the difference among codes in density-temperature PDF at the high-density end, and at the turnaround density from 10^4 K to lower temperatures. We have two hypotheses: (i) There is an intrinsic, uncontrolled difference on how codes deal with the self-shielding when using Grackle with primordial_chemistry=0, (ii) This is simply a consequence of different feedback recipes adopted in each code. To test our hypotheses, we decided to start a short run (only down to z=8) without star formation and feedback as soon as possible. Depending on the result, we will need to either (i) get deeper into the codes to find where the difference in the cooling routines is, or, (ii) just add a section in Paper "CosmoRun" describing how different feedback recipes affect the gas distribution in the density-temperature PDF. We all agree that a figure from the SF/feedback-free run in the Paper will be useful. Please submit the snapshot from this run, as soon as you get it. (3-2) Second, the differences between the minimum timesteps in AMR vs. SPH codes. Minimum timestep in SPH codes is an order of magnitude lower than the one in AMR. This is somewhat expected as AMR codes tend to refine later than SPH codes, especially at early times (e.g., O'Shea et al. 2005), resulting in less substructures in AMR runs at high-z. To test our hypothesis we will plot the minimum hydrodynamic softening length in SPH codes vs. twice the minimum cell size in AMR codes. Nonetheless, we decided to keep running the codes in the current configuration, but just add a description of this difference in Paper "CosmoRun". – (4) As of 11/2019, we also decided to run a short run (only down to z=7 or 7.5) with star formation but without stellar feedback as soon as possible, to test a new star formation threshold n_thres = 1H/cm^3 instead of 10H/cm^3, and check if different implementations of star formation (i.e., deterministic vs stochastic) are not driving the difference in star formation history. The test results will be shared via email; relevant figures in the Paper "CosmoRun" will be very useful. • Code Notes (as of mid-2019 to early-2020): – (1) ART-I: An issue with the refinement criteria (producing too many cells) has been identified and tackled. – (2) ENZO: Issues with the metallicity unit, and radiation background has been identified; a new run on its way. – (3) GADGET-3: An issue with the initial metallicity (10^{-4} Zs) has been identified and resolved. Nagamine is looking into the shortage of high-density/low-temperature gas. – (4) GEAR: An issue with parameter selection has been resolved. Revaz and Hausammann found and fixed an issue in Grackle regarding the high-z era cooling before UVB is activated. Check "Common physics: Cooling & UVB" above about which Grackle version we decided to use. – (5) GIZMO: Tested various choices of feebback to acquire better stellar mass. – (6) RAMSES: Issues with self-shielding and metallicity have been identified and fixed. A new feedback recipe including mechanical channel is being tested. – (7) Trident: Roca-Fabrega is making sure that Trident reads in all RAMSES metal species. Nagamine and Strawn are making sure that Trident reads in all GADGET metal species. Nagamine and Strawn are also trying to identify and fix bugs in Trident when dealing with different sizes of lines of sight to obtain column densities.

(4) Production Runs

	Data Locations, Visualizations, Metrics to Compare